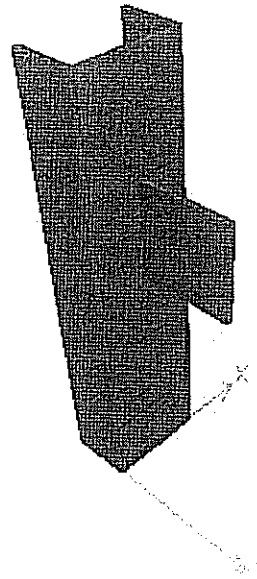
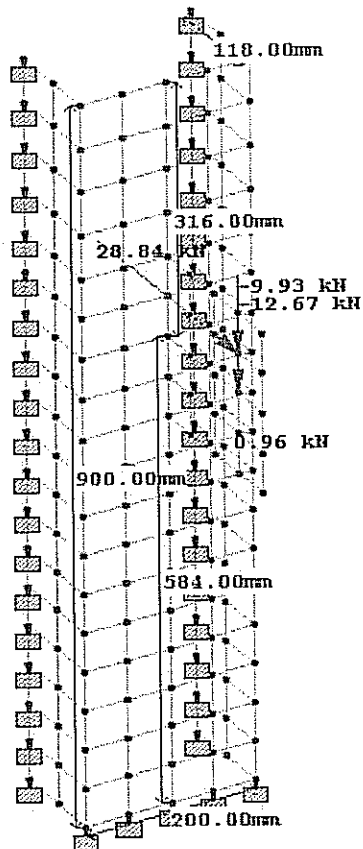


M.4 CHECK BASE PLATE BY FEA

M.4.1 GEOMETRIC MODEL AND LOAD DIAGRAM: BR-05



Model from Finite Element Analysis

M.4.2 STAAD INPUT AND OUTPUT DATA : for Finite Element Analysis of Bracket BR-05

```
*****
*          STAAD.Pro          *
*      Version 2007   Build 01   *
*  Proprietary Program of      *
*  Research Engineers, Intl.    *
*      Date=   AUG 7, 2008      *
*      Time=   11: 8:57        *
*      USER ID: ENGINEERING    *
*****
```

1. STAAD SPACE
- INPUT FILE: BR-05 BRACKETNEW.STD
2. START JOB INFORMATION
3. ENGINEER DATE 04-AUG-08
4. JOB NAME FRONT BRACKET
5. JOB CLIENT DMP (RASHIDIYA)
6. JOB NO 001
7. JOB REV 1
8. ENGINEER NAME AA

9. END JOB INFORMATION

10. INPUT WIDTH 79

11. UNIT MMS NEWTON

12. JOINT COORDINATES

13. 1 0 0 0; 2 0 484 0; 3 0 684 0; 4 0 900 0; 5 200 0 0; 6 200 484 0; 7 200 684 0
14. 8 200 900 0; 9 0 0 -118; 10 0 484 -118; 11 0 684 -118; 12 0 900 -118
15. 13 200 0 -118; 14 200 484 -118; 15 200 684 -118; 16 200 900 -118; 17 0 584 0
16. 18 200 584 0; 19 0 584 -118; 20 200 584 -118; 21 100 0 0; 22 100 900 0
17. 23 100 484 0; 24 100 684 0; 25 100 584 0; 26 100 484 195; 27 100 684 195
18. 28 100 584 195; 29 100 484 144; 30 100 684 144; 31 100 584 144; 32 100 634 195
19. 33 100 634 144; 34 100 534 195; 35 100 534 144; 36 0 0 -59; 37 0 48.4 -59
20. 38 0 48.4 -118; 39 0 48.4 0; 40 0 96.8 -59; 41 0 96.8 -118; 42 0 96.8 0
21. 43 0 145.2 -59; 44 0 145.2 -118; 45 0 145.2 0; 46 0 193.6 -59; 47 0 193.6 -118
22. 48 0 193.6 0; 49 0 242 -59; 50 0 242 -118; 51 0 242 0; 52 0 290.4 -59
23. 53 0 290.4 -118; 54 0 290.4 0; 55 0 338.8 -59; 56 0 338.8 -118; 57 0 338.8 0
24. 58 0 387.2 -59; 59 0 387.2 -118; 60 0 387.2 0; 61 0 435.6 -59; 62 0 435.6 -118
25. 63 0 435.6 0; 64 0 484 -59; 65 0 534 -59; 66 0 534 -118; 67 0 534 0
26. 68 0 584 -59; 69 0 634 -59; 70 0 634 -118; 71 0 634 0; 72 0 684 -59
27. 73 0 738 -59; 74 0 738 -118; 75 0 738 0; 76 0 792 -59; 77 0 792 -118
28. 78 0 792 0; 79 0 846 -59; 80 0 846 -118; 81 0 846 0; 82 0 900 -59
29. 83 200 0 -59; 84 200 48.4 -59; 85 200 48.4 0; 86 200 48.4 -118
30. 87 200 96.8 -59; 88 200 96.8 0; 89 200 96.8 -118; 90 200 145.2 -59
31. 91 200 145.2 0; 92 200 145.2 -118; 93 200 193.6 -59; 94 200 193.6 0
32. 95 200 193.6 -118; 96 200 242 -59; 97 200 242 0; 98 200 242 -118
33. 99 200 290.4 -59; 100 200 290.4 0; 101 200 290.4 -118; 102 200 338.8 -59
34. 103 200 338.8 0; 104 200 338.8 -118; 105 200 387.2 -59; 106 200 387.2 0
35. 107 200 387.2 -118; 108 200 435.6 -59; 109 200 435.6 0; 110 200 435.6 -118
36. 111 200 484 -59; 112 200 534 -59; 113 200 534 0; 114 200 534 -118
37. 115 200 584 -59; 116 200 634 -59; 117 200 634 0; 118 200 634 -118
38. 119 200 684 -59; 120 200 738 -59; 121 200 738 0; 122 200 738 -118
39. 123 200 792 -59; 124 200 792 0; 125 200 792 -118; 126 200 846 -59
40. 127 200 846 0; 128 200 846 -118; 129 200 900 -59; 130 150 0 0; 131 150 48.4 0
41. 132 100 48.4 0; 133 150 96.8 0; 134 100 96.8 0; 135 150 145.2 0
42. 136 100 145.2 0; 137 150 193.6 0; 138 100 193.6 0; 139 150 242 0
43. 140 100 242 0; 141 150 290.4 0; 142 100 290.4 0; 143 150 338.8 0
44. 144 100 338.8 0; 145 150 387.2 0; 146 100 387.2 0; 147 150 435.6 0
45. 148 100 435.6 0; 149 150 484 0; 150 150 534 0; 151 100 534 0; 152 150 584 0
46. 153 150 634 0; 154 100 634 0; 155 150 684 0; 156 150 738 0; 157 100 738 0
47. 158 150 792 0; 159 100 792 0; 160 150 846 0; 161 100 846 0; 162 150 900 0
48. 163 50 0 0; 164 50 48.4 0; 165 50 96.8 0; 166 50 145.2 0; 167 50 193.6 0
49. 168 50 242 0; 169 50 290.4 0; 170 50 338.8 0; 171 50 387.2 0; 172 50 435.6 0
50. 173 50 484 0; 174 50 534 0; 175 50 584 0; 176 50 634 0; 177 50 684 0
51. 178 50 738 0; 179 50 792 0; 180 50 846 0; 181 50 900 0; 182 100 484 48
52. 183 100 534 48; 184 100 484 96; 185 100 534 96; 186 100 584 48; 187 100 584 96
53. 188 100 634 48; 189 100 634 96; 190 100 684 48; 191 100 684 96
54. ELEMENT INCIDENCES SHELL
55. 1 9 36 37 38; 2 36 1 39 37; 3 38 37 40 41; 4 37 39 42 40; 5 41 40 43 44
56. 6 40 42 45 43; 7 44 43 46 47; 8 43 45 48 46; 9 47 46 49 50; 10 46 48 51 49
57. 11 50 49 52 53; 12 49 51 54 52; 13 53 52 55 56; 14 52 54 57 55; 15 56 55 58 59
58. 16 55 57 60 58; 17 59 58 61 62; 18 58 60 63 61; 19 62 61 64 10; 20 61 63 2 64
59. 21 10 64 65 66; 22 64 2 67 65; 23 66 65 68 19; 24 65 67 17 68; 25 19 68 69 70
60. 26 68 17 71 69; 27 70 69 72 11; 28 69 71 3 72; 29 11 72 73 74; 30 72 3 75 73
61. 31 74 73 76 77; 32 73 75 78 76; 33 77 76 79 80; 34 76 78 81 79; 35 80 79 82 12
62. 36 79 81 4 82; 37 5 83 84 85; 38 83 13 86 84; 39 85 84 87 88; 40 84 86 89 87
63. 41 88 87 90 91; 42 87 89 92 90; 43 91 90 93 94; 44 90 92 95 93; 45 94 93 96 97
64. ~~46 93 95 98 96; 47 97 96 99 100; 48 96 98 101 99; 49 100 99 102 103~~
65. ~~50 99 101 104 102; 51 103 102 105 106; 52 102 104 107 105; 53 106 105 108 109~~
66. 54 105 107 110 108; 55 109 108 111 6; 56 108 110 14 111; 57 6 111 112 113
67. 58 111 14 114 112; 59 113 112 115 18; 60 112 114 20 115; 61 18 115 116 117
68. 62 115 20 118 116; 63 117 116 119 7; 64 116 118 15 119; 65 7 119 120 121

69. 66 119 15 122 120; 67 121 120 123 124; 68 120 122 125 123; 69 124 123 126 127
70. 70 123 125 128 126; 71 127 126 129 8; 72 126 128 16 129; 73 21 130 131 132
71. 74 130 5 85 131; 75 132 131 133 134; 76 131 85 88 133; 77 134 133 135 136
72. 78 133 88 91 135; 79 136 135 137 138; 80 135 91 94 137; 81 138 137 139 140
73. 82 137 94 97 139; 83 140 139 141 142; 84 139 97 100 141; 85 142 141 143 144
74. 86 141 100 103 143; 87 144 143 145 146; 88 143 103 106 145; 89 146 145 147 148
75. 90 145 106 109 147; 91 148 147 149 23; 92 147 109 6 149; 93 23 149 150 151
76. 94 149 6 113 150; 95 151 150 152 25; 96 150 113 18 152; 97 25 152 153 154
77. 98 152 18 117 153; 99 154 153 155 24; 100 153 117 7 155; 101 24 155 156 157
78. 102 155 7 121 156; 103 157 156 158 159; 104 156 121 124 158
79. 105 159 158 160 161; 106 158 124 127 160; 107 161 160 162 22
80. 108 160 127 8 162; 109 1 163 164 39; 110 163 21 132 164; 111 39 164 165 42
81. 112 164 132 134 165; 113 42 165 166 45; 114 165 134 136 166; 115 45 166 167 48
82. 116 166 136 138 167; 117 48 167 168 51; 118 167 138 140 168; 119 51 168 169 54
83. 120 168 140 142 169; 121 54 169 170 57; 122 169 142 144 170; 123 57 170 171 60
84. 124 170 144 146 171; 125 60 171 172 63; 126 171 146 148 172; 127 63 172 173 2
85. 128 172 148 23 173; 129 2 173 174 67; 130 173 23 151 174; 131 67 174 175 17
86. 132 174 151 25 175; 133 17 175 176 71; 134 175 25 154 176; 135 71 176 177 3
87. 136 176 154 24 177; 137 3 177 178 75; 138 177 24 157 178; 139 75 178 179 78
88. 140 178 157 159 179; 141 78 179 180 81; 142 179 159 161 180; 143 81 180 181 4
89. 144 180 161 22 181; 145 23 182 183 151; 146 182 184 185 183; 147 184 29 35 185
90. 148 151 183 186 25; 149 183 185 187 186; 150 185 35 31 187; 151 25 186 188 154
91. 152 186 187 189 188; 153 187 31 33 189; 154 154 188 190 24
92. 155 188 189 191 190; 156 189 33 30 191; 157 29 26 34 35; 158 35 34 28 31
93. 159 31 28 32 33; 160 33 32 27 30
94. ELEMENT PROPERTY
95. 1 TO 160 THICKNESS 12
96. DEFINE MATERIAL START
97. ISOTROPIC STEEL
98. E 205000
99. POISSON 0.3
100. DENSITY 7.68195E-005
101. ALPHA 1.2E-005
102. DAMP 0.03
103. END DEFINE MATERIAL
104. CONSTANTS
105. MATERIAL STEEL ALL
106. SUPPORTS
107. 1 5 9 TO 16 19 TO 21 38 41 44 47 50 53 56 59 62 66 70 74 77 80 86 89 92 95 -
108. 98 101 104 107 110 114 118 122 125 128 130 163 FIXED
109. LOAD 1 LOADTYPE NONE TITLE LOAD CASE 1
110. JOINT LOAD
111. 31 FY -12670
112. 35 FZ 960
113. 33 FZ 28840
114. 33 FY -9930
115. PERFORM ANALYSIS
116. PRINT MAX. ELEMENT STRESS